

THE THEORY OF MEASURES AND INTEGRATION

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ABSTRACT. Lebesgue's integral patches up many of the defects of Riemann's integral, at the expense of the need to develop a notion of *measure*. These notes offer a broad and accelerated overview of the Lebesgue integral and measure theory with several exercises, open ended questions, and examples.

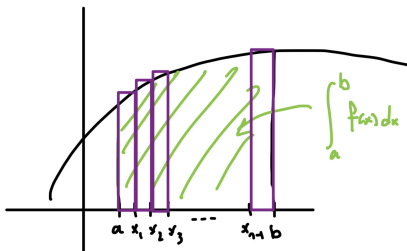
1. LECTURE 1: SOME MOTIVATION, BASIC DEFINITIONS AND RESULTS

1.1. Motivation. The notions of length, area, and volume are so fundamental to our experience of the world, that they are some of the first concepts that are taught in school. Length is at the forefront of our minds when we are trying to get somewhere. Not only a physical length in terms of distance traveled, but also a length of the stretch of time we will need to get there. We encounter volume on a daily basis when we drink from a bottle of water, or fill up a tank of gas. Area is no exception, can you think of some daily encounters you have with area?

As intuitive as these concepts are, under a mathematical microscope, they are a little nebulous. A pure mathematician might look at this vague intuitive idea and see a gap in our mathematical understanding. This gap has not been filled by standard education, because the details can be surprisingly subtle.

Okay but why do we care? Is pinning down the mathematics of length just a little puzzle? Is it just a mathematical curiosity? On a practical level, you aren't going to be drinking water any differently after understanding the notion of volume mathematically. However, the gap in understanding of these notions turn out to be a significant gap in understanding other mathematical ideas. For example, few people would dispute the ubiquitousness of calculus in modeling everything from physics to epidemiology to market trends. However, at the heart of calculus is the same idea of length in the definition of the *integral*. Let's take a closer look.

The heart of Riemann's approach is to chop up an area that's tough to understand, in terms of simpler boxes. This idea is at least as old as Eudoxus, around 400 BCE. However, Riemann had a very specific approach to doing this. His idea was to chop up the domain of a function and construct rectangular areas, corresponding to the height of the function in those boxes, as in the following image:



Then we could write the approximation:

$$\int_a^b f(x)dx \approx \sum_{k=1}^n f(x_k^*)(x_k - x_{k-1}),$$

which is simply the sum of rectangular areas we have made. Here is where the notion of length gets inserted, for each summand $f(x_k^*)(x_k - x_{k-1})$ is the product *height* $\times$ *length* of a rectangular area.

This seems all fine and dandy; length seems explicit and leads to the areas we would expect. However, there turn out to be significant holes in Riemann's approach, due to the rigidity in our idea of length. For example, The class of functions that we can integrate via Riemann's approach is not particularly well defined (in the sense of explicit characteristics of the functions themselves) or well behaved. This leads to things like being able to escape our space with a relatively tame sequence.

To get a feel for this, consider the following: if $f = 0$ then $\int_0^1 f(x)dx = 0$ (let's restrict our domain of interest to $[0, 1]$ for this discussion). So far, so trivial. But you may know from calculus, that if f is changed at a point, or even finitely many points, this does not change its integral, for there is no area under a point. So, for example, let's pick points $x_1, x_2, \dots \in [0, 1]$ and define

$$f_n(x) = \begin{cases} 1 & \text{if } x \in \{x_1, \dots, x_n\} \\ 0 & \text{otherwise} \end{cases}.$$

Then $\int_0^1 f_n(x)dx = 0$. Notice that $f_{n+1} \geq f_n$ for all $n \in \mathbb{N}$, and if say, $x_n = \frac{1}{n}$, then

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1 & \text{if } x = \frac{1}{k}, k \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

is Riemann integrable with integral zero (check this). But if we just select some different points, we run into some trouble. For example, if instead the x_n 's enumerate $\mathbb{Q} \cap [0, 1]$, then we get

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0, 1] \\ 0 & \text{otherwise} \end{cases}.$$

It is a nice exercise to show that this is simply not Riemann integrable, for in every interval there are rational numbers and irrational numbers.

So on one hand, we should expect that this function is integrable with integral zero, but on the other hand, our approach to the integral inhibits us from making sense of it! In more analytical language, we could have defined a (semi)norm corresponding to the integrals of these functions, for which they form a Cauchy sequence, yet their limit does not exist!

While this seems like a minor problem to have, this lack of *completeness* of the space of integrable functions has inhibited a lot of powerful integration tools, such as a simple monotone convergence theorem (counter example seen above).

These holes plagued integration theory until Henri Lebesgue patched them up. Lebesgue sat down and inspected the notion of length much closer. In 1901 he came out with his theory of measures, which not only gets a good handle on the notion of length, but also opens up a whole new set of objects, which can be thought of as *generalized lengths*, to be studied. This laid the groundwork for his 1902 paper on the integral. His idea of the integral was a brilliant reformation of Riemann's idea. In fact, it was as simple as looking at the integral in a different way: chop up the range instead of the domain! This approach is not quite so simple, for there was a tradeoff. Instead of being simply defined like Riemann's integrable, a price had to be paid; we needed a more robust notion of length.

It is this theory of measure and integration that we study here. We begin with looking at the concepts of length, area, and volume "under a microscope", seeing where we can capture its essence. This leads to connections and broader definitions, which beget their own behaviours and properties. In Lecture 2, we use this notion of measure to look at Lebesgue's approach to the integral. We mention several very nice limit theorems that result, and how the Lebesgue integral extends Riemann's. In Lecture 3, we take a detour and think of integrals themselves as linear functionals and measures as distributions. This opens up a bridge to studying this topic through the lens of functional analysis and operator algebras. Lecture 4 discusses how we can compare measures, and in some cases, transform them into each other through the use of *densities*. Lecture 5 asks the question "what if lengths could be negative?". Far from being an idle curiosity, this question too seems to fit right in with much of the theory developed to that point. In Lecture 6,

we finally arrive at a treatment of area and volume in the form of *product measures*. These are a natural (and categorical) construction that bridges length, area, and volume. Lecture 7 deals with the widely useful and powerful fundamental theorem of calculus. In particular, we saw how Riemann's integral has some holes in it, so we discuss the extra power we get from Lebesgue's approach.

Length, Area, Volume, and Probability

We start with a simple question of *what is length?* We should be able to get some sort of mathematical description for it. The standard objects of length measurements are one-dimensional, so it is fair to restrict ourselves to working in $\mathbb{R}$. In this setting, the objects of interest are, at their simplest, line segments, which are in turn naturally modeled by *intervals*.

So if we have an interval $[a, b] \subseteq \mathbb{R}$, it is then quite immediate to suggest its length:

$$\text{length of } [a, b] = b - a.$$

We could end our inquiry here, but surely we can extend this notion of length to more than just intervals? For one, disjoint unions of intervals still have a reasonable length associated, by just adding up their lengths:

$$\text{length of } [a, b] \sqcup [c, d] = (b - a) + (d - c).$$

In fact, even in the case of a not disjoint union, we would still simply get a larger interval, for which we could measure the length!

By now, you likely have the idea that what we mean by a *length* is a function ℓ that maps subsets of $\mathbb{R}$ to non-negative real numbers, satisfying $\ell([a, b]) = b - a$ for any interval $[a, b]$. One question that immediately presents itself then is: can we find the length of every subset of $\mathbb{R}$? That is, is the domain of ℓ all of $\mathcal{P}(\mathbb{R})$? In a sort of trivial way, we can find a function satisfying the above for all sets (simply take all sets that aren't intervals or disjoint unions of intervals to have "length" 0). But this is far from a satisfying definition of length. So what properties might we want?

Let's try to pin down some reasonable properties. The first property that we want is to be able to find the length of any subset. Next, given that our definition above left us with some pathological examples of lengths, let's upgrade it to be *countably additive* (as opposed to just *finitely additive*). That is, if $\{A_k\}_{k \in \mathbb{N}}$ is a family of pairwise disjoint sets, we demand that

$$\ell\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \sum_{k=1}^{\infty} \ell(A_k).$$

Lastly, let's impose a reasonable symmetry. For example, the length of a set should not depend on *where* the set is located, only its *content*. So we demand further that for any real number r and subset S , we have

$$\ell(S + r) = \ell(S).$$

This property is known as *translation invariance*.

These three properties seem quite reasonable. There's just one problem. There is no such function! Somewhat strangely, there exist sufficiently strange subsets of $\mathbb{R}$, known as *Vitali sets*, which simply break this attempt at a definition. Let's show this now.

Theorem 1.1 (Vitali). *There does not exist a positive, countably additive, translation invariant set function ℓ on $\mathcal{P}(\mathbb{R})$ such that $\ell([a, b]) = b - a$ for all intervals $[a, b]$.*

Proof. The rough idea of Vitali's sets are that they have the property that a countably infinite number of their translates union, disjointly, to cover the interval $[0, 1]$, but stay bounded. This leaves us with the contradictory conclusion that a countable sum of the same number can be a finite positive number.

The construction is as follows. First, say $x \sim y$ if $x - y \in \mathbb{Q}$. It is not hard to see that $\sim$ is an equivalence relation on $[0, 1]$. Let V be a set of representatives chosen, by the Axiom of Choice, one from each equivalence class. We claim that V creates a contradiction.

First note that different translates are disjoint. In particular, if $r_1, r_2 \in \mathbb{Q} \cap [-1, 1]$ and $r_1 \neq r_2$, then

$$(V + r_1) \cap (V + r_2) = \emptyset.$$

To see this, suppose for contradiction that there is some x in both sets. Then by definition of V , there would be some representatives y_1 and y_2 such that $x = y_1 + r_1 = y_2 + r_2$. This leads to the conclusion that $y_1 - y_2 \in \mathbb{Q}$, which either contradicts the fact that $r_1 = r_2$ (in the case when $y_1 = y_2$) or the fact that there is exactly one representative from each equivalence class in V .

Next, if $\{r_1, r_2, \dots\}$ is an enumeration of $\mathbb{Q} \cap [-1, 1]$, then

$$[0, 1] \subseteq \bigcup_{n=1}^{\infty} (V + r_n) \subseteq [-1, 2].$$

The first inclusion is readily seen from the fact that any element of $[0, 1]$ is equivalent to one element of V with difference r_n for some n . The second inclusion is just noting that $V \subseteq [0, 1]$.

Lastly, we compute.

$$\begin{aligned} \ell \left(\bigcup_{n=1}^{\infty} (V + r_n) \right) &= \sum_{n=1}^{\infty} \ell(V + r_n) && \text{(countable additivity)} \\ &= \sum_{n=1}^{\infty} \ell(V) && \text{(translation invariance)} \end{aligned}$$

Furthermore, it must be the case that ℓ is *monotone*: if $A \subseteq B$ then $\ell(A) \leq \ell(B)$. This is because we could write $B = A \sqcup B \setminus A$ and use the additivity and positivity of ℓ . Thus since $[0, 1] \subseteq \bigcup_{n=1}^{\infty} (V + r_n) \subseteq [-1, 2]$, we must have by monotonicity that

$$1 = \ell([0, 1]) \leq \ell \left(\bigcup_{n=1}^{\infty} (V + r_n) \right) \leq \ell([-1, 2]) = 3$$

But this leaves us in an impossible situation. If $\ell(V) = 0$, then we get $1 \leq 0$. If $\ell(V) > 0$, then we get $\infty \leq 3$. $\square$

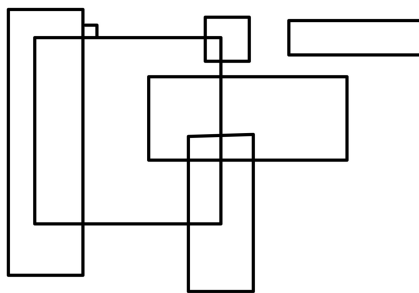
The existence of Vitali sets forces us to give up one of our three desired properties. So which one do we give up? Well, the translation invariance feels bad to give up, for how could one's measuring stick change in length by moving around? (Strictly speaking, by our demand that the lengths of intervals are what we expect, a measuring stick would not change its length, but dropping translation invariance may, a priori, lead to sets having different lengths depending on where they are located.) Maybe we should give up the countable additivity?

Let's pause here for a short digression. Let's now ask the simple question of what is area? If you are inspired from the discussion above about lengths, then you might aim at definition of area as a function which maps subsets of $\mathbb{R}^2$ into the non-negative reals satisfying some sort of additivity, and that they measure certain simple subsets "properly". And this is indeed how one can define area! The easiest such sets are the *boxes*, which take the form $[a, b] \times [c, d] \subseteq \mathbb{R}^2$. It is natural to suggest then that the area of such a box be:

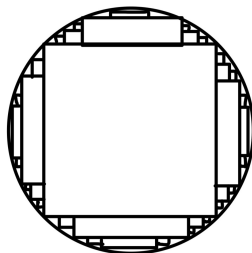
$$A([a, b] \times [c, d]) = (b - a)(d - c).$$

One may note the direct connection this has with the length function in expression above. This will be addressed explicitly in Lecture 6.

But as we saw before, we ought not be too greedy in demanding properties. Translation invariance still seems like a must. So again we raise the question: do we get rid of countable additivity? If so, then we can only reasonably control the areas of *finite* unions of boxes. This allows us to make all kinds of shapes like the following.



But one thing obviously lacking from these pictures is any sense of curvature. Shapes that aren't "boxy" enough are out of our control if we do not accept countable additivity! Thus shapes like circles (or even triangles!) cannot be reasonably controlled in terms of their area if we do not accept countable additivity. On the other hand however, accepting countable additivity does allow us to find areas of circles and triangles in the following way:



So instead, we accept countable additivity and translation invariance, which leaves only one property to reject: the ability to measure all possible subsets. This feels bad as this was what we set out to do in the first place, but the cost of getting what we want here is simply too high to be reasonable. So we can only measure some family of subsets of $\mathbb{R}$. What family can this be?

Well we saw how we wanted countable additivity. This means we ought to be able to take countable unions of any measurable sets, and have them stay measurable. It is reasonable also to demand a certain *normalizing* that the empty set has no length (or area), so we include the empty set. And lastly, one might notice that when we took finite unions of intervals, their complements turned out also to be finite unions of intervals! So let us also tack on the property that if S is a set that can be measured, then we demand that $S^c = \mathbb{R} \setminus S$ can be measured as well.

One might wince at our greediness with that last property, however, luckily this time there is no issue. There is a naturalness to it as well, for if we have some measurable S which is *bounded* so that $S \subseteq [-n, n]$ for some $n \in \mathbb{N}$, then the length of $[-n, n] \setminus S$ must naturally be $2n - \ell(S)$. So we readily infer the measures of complements of sets, at least in their boundedness (more on this later, when talking about σ -finiteness).

A collection of subsets of $\mathbb{R}$ satisfying the three properties above (empty set, closure under countable unions, and closure under complements) is known as a σ -algebra. One may note however, that this is easily abstracted beyond $\mathbb{R}$ or $\mathbb{R}^2$.

Before we jump into the formal definitions, one might make the observation that the properties of lengths and areas as well as the types of sets that we work with show up in a seemingly separate branch of mathematics: *probability theory*. Indeed, instead of thinking of subsets as things we need to assign lengths or areas to, we can think of subsets as *events* that we assign *probabilities* to.

In this way, the collection of events naturally has the same σ -algebra structure, but this time on a *sample space* Ω (as opposed to $\mathbb{R}$). Indeed, we should always have the *null event*, $\emptyset$, and the *universal event*, Ω . For any event E , we should have its *negative-event* (that is if E is an event, then "not E " is equally valid as an event), which corresponds to $E^c = \Omega \setminus E$. And lastly, for any two events E, F , we should have their *or-event* (that is if E and F are events, then we could naturally ask for " E or F "), which corresponds to $E \cup F$. Now this is closure under countable unions, but it still makes good sense to upgrade this to countable unions (in fact, area and length

measures themselves are sometimes probability measures). So this leaves us to conclude that the set of events that we can assign probabilities to form a σ -algebra!

It is also not too hard to reason that the two properties we wanted for our length and area measures also extend to probability measures. The null event, $\emptyset$, should have probability zero. And if we have a (countable) collection of pairwise disjoint events, then the probability of their union should be the sum of their individual probabilities. So probability functions somehow have the same character as length or area functions.

We will discuss probability theory more at length later on, but for now, let us begin recording our observations thus far, beginning with a σ -algebra, which captures the types of subsets we should reasonably be able to measure.

Definition 1.2. Given a set Ω , a σ -algebra of subsets of Ω is a family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that

- (i) $\emptyset, \Omega \in \mathcal{F}$,
- (ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and
- (iii) if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ then $\bigcup_{k \in \mathbb{N}} A_k \in \mathcal{F}$.

We'll say a pair $(\Omega, \mathcal{F})$ is a *measurable space*.

One might ask why there is a bias for unions here. For if we take our probabilistic interpretation that unions are *or-events*, then what about the corresponding *and-events*? Well, it turns out that these also come for free from our definition. (It is largely a matter of taste or style, but one shared fairly universally, that we don't include redundant information in our definitions.)

Question 1. Show that if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ then

$$\bigcap_{k \in \mathbb{N}} A_k \in \mathcal{F}.$$

Let's look at a few examples of σ -algebras now.

Example 1.3. Beginning with a few simple examples. If Ω is a set, then $\{\emptyset, \Omega\}$ is a σ -algebra and $\mathcal{P}(\Omega)$ (the collection of all subsets of Ω) is a σ -algebra.

Actually *all* finite σ -algebras are essentially just power sets. (See Question 12)

But it would be dull if all σ -algebras were just power sets. The most useful ones can be surprisingly slippery to define. We usually want to specify some collection of sets that we know we can measure (e.g. the intervals) which is not a σ -algebra, and be able to find the smallest σ -algebra that contains that collection. This allows us to extend our unstructured family into a σ -algebra acting as a natural domain for the measure that sits on top.

Definition 1.4. Given a set Ω and a family of subsets $\mathcal{S} \subseteq \mathcal{P}(\Omega)$, we define the σ -algebra generated by $\mathcal{S}$ to be the smallest σ -algebra containing $\mathcal{S}$. That is,

$$\sigma(\mathcal{S}) := \bigcap_{\substack{\mathcal{F} \subseteq \mathcal{P}(\Omega) \\ \mathcal{F} \text{ is a } \sigma\text{-algebra} \\ \mathcal{S} \subseteq \mathcal{F}}} \mathcal{F}.$$

It is a simple exercise to show that these generated σ -algebras are indeed σ -algebras. As alluded to before, we could let $\mathcal{S}$ be the collection of all intervals say, or if our considerations lean towards topology and continuity, then we might want to let $\mathcal{S}$ be a family of open sets!

Example 1.5. Let (X, τ) be a topological space. Then the σ -algebra generated by the topology (i.e. all the open sets), *Borel σ -algebra*, which we will denote by

$$\mathcal{B}(X) := \sigma(\tau).$$

For example, take $\mathbb{R}$ with its usual (metric) topology. Then the Borel σ -algebra is a non-trivial (i.e. not $\mathcal{P}(\mathbb{R})$) σ -algebra containing all open subsets of $\mathbb{R}$. We will revisit this non-triviality after constructing the Lebesgue measure. In fact, it will be seen that sets like the Vitali set which is constructed as a counter example to our ability to measure the lengths of all possibly subsets of $\mathbb{R}$ (see Theorem 1.1), cannot be Borel.

Question 2. Show that $\mathcal{B}(X)$ contains all closed sets, all F_σ sets, and all G_δ sets. (From topology: A set is closed if it is the complement of an open set, it is F_σ if it can be written as a countable union of closed sets, it is G_δ if it can be written as a countable intersection of open sets.)

Curiosity 1. What is the cardinality of $\mathcal{B}(\mathbb{R})$? Seems like it contains a lot of sets, so it could be that $|\mathcal{B}(\mathbb{R})| = |\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}}$.

However, it definitely does not contain all sets. So perhaps we have $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$?

If the latter is the case, that would be quite shocking, as all imaginable or constructable sets appear to be Borel, yet there is a whole size of infinity larger than the Borel sets composed of the unimaginable ones!

Question 3. Show that Borel subsets of $\mathbb{R}$ (with its metric topology) are also generated by the following collections

- (i) $\{(a, b] : -\infty \leq a < b < \infty\}$,
- (ii) $\{(a, \infty) : a \in \mathbb{R}\}$
- (iii) $\{(a, b) : a < b \in \mathbb{R}\}$

Curiosity 2. As a passing curiosity about σ -algebras, to help one understand them in a slightly different way, one might ask about how many elements they can contain.

It turns out that all the finite σ -algebras are order-isomorphic to a powerset (see Question 12). So the only finite sizes for σ algebras are powers of 2.

But what about infinite ones? Can an infinite σ -algebra be countable? This is an accessible, but not necessarily easy, problem. No need to get hung up on it now (unless you want to), but it is left as an exercise at the end of this lecture.

Now that we have the types of domains for our measures set up, we can formally define our main objects of interest.

Definition 1.6. Given a measurable space $(\Omega, \mathcal{F})$, a (positive) *measure* on $\mathcal{F}$ (or on Ω if $\mathcal{F}$ is understood) is a function $\mu : \mathcal{F} \rightarrow [0, \infty]$ such that

- (i) $\mu(\emptyset) = 0$, and
- (ii) if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

A triple $(\Omega, \mathcal{F}, \mu)$ where $\mathcal{F}$ is a σ -algebra of subsets of Ω and μ is a measure on $\mathcal{F}$ will be called a *measure space*.

Example 1.7. The first example that we care about, given our discussion at the beginning of these notes is the *Lebesgue measure* λ . As we saw before, its domain cannot be all of $\mathcal{P}(\mathbb{R})$, so it needs to be restricted. It does have $\mathcal{B}(\mathbb{R})$ as a subset of its domain, so for now we will think of it as the mapping $\lambda : \mathcal{B}(\mathbb{R}) \rightarrow [0, \infty]$.

The Lebesgue measure has the following properties; first, for any interval $[a, b]$ we have

$$\lambda([a, b]) = b - a,$$

and secondly, if r is any real number, and S is any Borel set, then

$$\lambda(S + r) = \lambda(S).$$

(Check that $S + r$ is still a Borel set if S is.) It turns out that these two properties are sufficient to determine the measure of any Borel subset of $\mathbb{R}$!

The proof that such a measure exists is deferred to a later chapter, as it relies on a technical and unenlightening *extension* process due to Caratheodory.

The area and volume measure actually don't require independent construction, rather they are built from the Lebesgue measure λ by taking *products*, as alluded to in our opening discussion. So in fact, the area and volume (Lebesgue) measures are often denoted

$$\lambda^2 = \lambda \times \lambda \quad \text{and} \quad \lambda^3 = \lambda \times \lambda \times \lambda.$$

We will discuss these extensions and constructions in Lecture 6.

Now, since the definition of measure extends far beyond length, area, and volume, there are plenty of other measures to explore. Some of these will be quite interesting when we discuss the integral in Lecture 2. Let's take a look at these now.

Example 1.8. Take $\Omega = \mathbb{N}$ and $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define the *counting measure* on $\mathbb{N}$, $\mu_{\text{count}} : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ by

$$\mu_{\text{count}}(S) = |S|.$$

It is not hard to see that this satisfies all requirements of our definition.

In fact, we can define this same counting measure on any set, which we may encounter later. But the counting measure on $\mathbb{N}$ will be of particular importance to us.

One may reasonably call this a zero-dimensional volume measure, like length is to one-dimension and area is to two-dimensions!

Example 1.9. An even simpler example is, for any Ω and any $a \in \Omega$, we have the *Dirac mass at a* measure, which is defined by

$$\delta_a(S) = \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{otherwise} \end{cases}.$$

If measures are showing us how volume or mass is distributed, then this one says that all of it is concentrated at the point a . It is a point-mass! More on this mass perspective in Lecture 3.

Dirac masses are some of the simplest measures we can have but they are surprisingly useful.

Question 4. *The two measures we saw above are connected in a very direct way. Show that*

$$\mu_{\text{count}} = \sum_{n=1}^{\infty} \delta_n.$$

Question 5. *Motivated from the previous question, show that adding two measures results in a measure.*

Not all linear combinations of measures are permissible. Which ones are?

Such a space is often called a convex cone.

Question 6. *Show that every measurable space can be upgraded to a measure space with some measure.*

Curiosity 3. *Can we classify the set of measures somehow? Maybe in the simplest setting, say $|\Omega| < \infty$. Can we characterize all measures on $(\Omega, \mathcal{P}(\Omega))$? In this setting, if $\mathcal{F}$ is a sub- σ -algebra of $\mathcal{P}(\Omega)$ and μ is a measure on $(\Omega, \mathcal{F})$, can μ be extended to $\mathcal{P}(\Omega)$ for free?*

Now that we have some examples to play with, we are free to ponder some general patterns that they all share. For example, one may expect that larger sets should have larger measures, as is intuitive from length, area, and probability. This is indeed true.

Lemma 1.10. *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $A, B \in \mathcal{F}$, with $A \subseteq B$.*

A proof is provided here in order to give a taste of these types of measure theoretic arguments, but it is not difficult so you are encouraged to try to come up with the argument yourself before reading it!

Proof. If $A \subseteq B$, then we may write $B = A \sqcup (B \setminus A)$. Note that $B \setminus A = B \cap A^c$, so that since $\mathcal{F}$ is a σ -algebra, it is closed under complements and countable intersections (how do we also get closure under finite intersections?), implying that $B \setminus A \in \mathcal{F}$.

Now countable (and hence finite, why?) additivity of μ lets us say that

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A). \quad \square$$

We can also poke at the definition a little bit. One may notice, for example, that the second property only allows us to handle measures of disjoint unions, whereas our σ -algebras are closed under potentially non-disjoint unions as well. However, this is no real limitation, as we can always *disjointify* a family, and at least get sub-additivity. This is left as an exercise, but it follows the rough ideas as the proof above.

Question 7. For a given family of sets $\{A_k\}_{k \in K}$, show that there is a disjoint family $\{B_k\}_{k \in K}$ such that

$$\bigcup_{k=1}^n A_k = \bigcup_{k=1}^n B_k.$$

Question 8. Show that, with the setup above,

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \sum_{k=1}^{\infty} \mu(B_k).$$

Argue that for any countable collection $\{A_k\}_{k \in K} \subseteq \mathcal{F}$, we have sub-additivity of measure:

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) \leq \sum_{k=1}^{\infty} \mu(A_k).$$

We also quickly get using the same ideas the following computational tool for dealing with non-disjoint sets:

Proposition 1.11 (Inclusion-Exclusion). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $A, B \in \mathcal{F}$. Then

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Question 9. Prove the Inclusion-Exclusion principle above. Note that this is a standard exercise for probability theory, and it greatly illuminated by a picture!

Remark 1.12. One other objection one might have to the definition of measure is that we can seemingly define it equally fine for just finite unions. For example, one can define a *finitely-additive measure* on an algebra $\mathcal{F}$ of subsets of Ω as a map $\mu : \mathcal{F} \rightarrow [0, \infty]$ satisfying

- (i) $\mu(\emptyset) = 0$, and
- (ii) If $A_1, \dots, A_n \in \mathcal{F}$ are pairwise disjoint, then

$$\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n).$$

It is immediate that all measures in our original sense are finitely additive in this sense (why?). So, what benefit do we get from allowing for infinite unions?

Aside from our discussion at the start of this lecture about countable additivity being appropriate for our area measures, a nice benefit of allowing countable infinities is that we can use the tools of limits and continuity, which vastly expands our scope. These kinds of arguments will be crucial for developing Lebesgue's integral later on.

The following is a result that is unique to countably-additive measures.

Theorem 1.13 (Continuity of Measure). If $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ with $A_k \subseteq A_{k+1}$ for all $k \in \mathbb{N}$ then

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Furthermore, if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ with $A_k \supseteq A_{k+1}$ for all $k \in \mathbb{N}$, and if $\mu(A_1) < \infty$, then

$$\mu \left(\bigcap_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Question 10. Prove the above result. (Hint: use a disjointification!) Also, show that without the assumption that $\mu(A_1) < \infty$ in the second part of the theorem above, the statement may not be true.

Yet another aspect of the definition that we may poke at is that we can potentially include infinite values for μ . On one hand, this may be surprising as we don't often deal with infinite values in math. A key obstruction in dealing with infinities directly is our lack of ability to do any algebra with them (this issue will present itself in Lecture 5). On the other hand, many subsets of $\mathbb{R}$ clearly must have infinite length ($\mathbb{R}$ itself for one).

On the other hand, if we consider probabilities, then our whole space must have *finite* measure. In fact, a probability measure is simply a measure such that $\mu(\Omega) = 1$. By monotonicity then, any $A \subseteq \Omega$ must have $\mu(A) \leq 1$.

Upon further reflection, if $0 < \mu(\Omega) < \infty$, then μ behaves very much like a probability measure; we can simply normalize it to a probability measure by letting

$$\tilde{\mu} = \frac{1}{\mu(\Omega)} \mu.$$

Let's establish some notation to better discuss these objects.

Notation 1.14. We highlight the following collections of measures.

- $\text{Meas}^+(\Omega, \mathcal{F}) := \{\mu : \mu \text{ is a (positive) measure on } (\Omega, \mathcal{F})\}$,
- $\text{Meas}_{\text{fin}}^+(\Omega, \mathcal{F}) := \{\mu \in \text{Meas}^+(\Omega, \mathcal{F}) : \mu(\Omega) < \infty\}$, and
- $\text{Prob}(\Omega, \mathcal{F}) := \{\mu \in \text{Meas}^+(\Omega, \mathcal{F}) : \mu(\Omega) = 1\}$.

With this notation, our discussion above says that

$$\text{Meas}_{\text{fin}}(\Omega, \mathcal{F}) = R^{\geq 0} \cdot \text{Prob}(\Omega, \mathcal{F}).$$

Curiosity 4. What structure(s) do these sets have? Any algebraic or topological ones? Well we noticed before that any positive linear combination of measures is still a measure, which carries over to $\text{Meas}_{\text{fin}}^+(\Omega, \mathcal{F})$. So $\text{Meas}_{\text{fin}}^+(\Omega, \mathcal{F})$ is a convex cone.

The same cannot quite be said about $\text{Prob}(\Omega, \mathcal{F})$, for scalar multiples of probability measures are almost certainly not probability measures. However, it's not too hard to show that convex combinations of probability measures are probability measures! This leaves us with the observation that $\text{Prob}(\Omega, \mathcal{F})$ is at least a convex set.

To get at some more topological properties about these collections, it is often simpler to work on a relatively well-behaved underlying space let when Ω is a compact metric space and $\mathcal{F}$ is the Borel σ -algebra on $\mathcal{F}$. We will discuss various aspects of these types of settings in future lectures. But see if you can think of some natural conditions yourself!

Ultimately the set of probability measures is a rich and fascinating space to explore. So much so, that it has an entire branch of mathematics devoted to its study: probability theory!

As great as finite measures are, we don't want to just toss out all the non-finite measures. This would exclude a wealth of useful examples, like our length (Lebesgue) measure on $\mathbb{R}$!

Infinite measures are not all the same either. Some of them are relatively well-behaved in the sense that they are built from finite ones. Take for example the length (Lebesgue) measure on $\mathbb{R}$. We can find the length of a set S by taking the limit:

$$\ell(S) = \lim_{n \rightarrow \infty} \ell(S \cap [-n, n]).$$

Definition 1.15. A measure on $(\Omega, \mathcal{F})$ is called σ -finite if there exists a countable collection of sets $\{S_n\}_{n=1}^{\infty} \subseteq \mathcal{F}$ such that $S_n \subseteq S_{n+1}$ for all $n \in \mathbb{N}$, $\bigcup_{n=1}^{\infty} S_n = \Omega$, and

$$\mu(S_n) < \infty.$$

So σ -finite measures are those measures which are approximately finite, or can be understood through finite measures in a special way. Of course all finite measures are σ -finite, but it is not true the other way around (e.g. the Lebesgue/length measure).

However not all measures can be broken down in this way. For example, one consequence of the definition is that every point must have finite measure. So σ -finite measure cannot be pathological in the sense of having infinite point-masses. However, there are still measures that aren't pathological in this way, but still fail to be approachable with finite measure sets.

Question 11. Find a measure space $(\Omega, \mathcal{F}, \mu)$ which is not σ -finite, yet $\mu(\{\omega\}) < \infty$ for all $\omega \in \Omega$.

This concept of σ -finiteness will reappear several times throughout our upcoming investigations.

1.2. Exercises.

Question 12. We will show that all finite measurable spaces are of the form $(\Omega, \mathcal{P}(\Omega'))$. Let $(\Omega, \mathcal{F})$ be a measurable space where $|\mathcal{F}| < \infty$.

- i Say a nonempty subset $A \in \mathcal{F}$ is minimal if $B \in \mathcal{F}$ and $B \subseteq A$ implies that $B = \emptyset$ or $B = A$. Show that the collection of minimal elements of $\mathcal{F}$ partitions Ω (i.e. they are all disjoint and union to Ω).
- ii Find an appropriate set Ω' and a function $f : \Omega \rightarrow \Omega'$ such that $\hat{f} : \mathcal{F} \rightarrow \mathcal{P}(\Omega')$, defined by $\hat{f}(A) := f(A)$ is an order-isomorphism of $\mathcal{F}$ to $\mathcal{P}(\Omega')$.

Question 13. Can infinite σ -algebras be countable? (Hint: $|\mathcal{P}(\mathbb{N})| > |\mathbb{N}|$.)

Question 14. Show that the σ -algebra generated by singletons on $\mathbb{R}$ is not $\mathcal{P}(\mathbb{R})$.

Question 15. Show that there are finitely-additive measures that are not countably-additive. (Hint: such a measure must fail the continuity theorem for some collection of sets)

Question 16. Show every measure on $\mathbb{N}$ is uniquely determined by a weight function $w : \mathbb{N} \rightarrow [0, \infty]$ in the sense that for any $S \subseteq \mathbb{N}$,

$$\mu(S) = \sum_{n \in S} w(n).$$

Deduce which measures are σ -finite.

Question 17. Show that if μ is σ -finite, and $\mu(A) = \infty$, then there is a $B \subsetneq A$ such that $\mu(B) = \infty$.
Need this be true if μ is not σ -finite?

2. LECTURE 2: LEBESGUE'S INTEGRAL

As we briefly discussed in Lecture 1, Riemann's integral was a very successful approach that has some annoying limitations.

Motivation: balancing act

Question 18. *If we want to have our cake and eat it too, what functions satisfy having every preimage of an interval being also an interval?*

2.1. Permissible Functions. To allow for the chaos in Lebesgue's idea for the integral, we need to demand a certain property of our functions. In particular, for f to be integrable, a necessary condition is that the pre-images of intervals must have the ability to be measured. This condition is called *measurability*.

Definition 2.1. Given a measure space $(\Omega, \mathcal{F}, \mu)$, a function $f : \Omega \rightarrow \mathbb{R}$ is *measurable* if for all $B \in \mathcal{B}(\mathbb{R})$,

$$f^{-1}(B) \in \mathcal{F}.$$

For convenience, we denote the set of measurable functions on Ω by $\text{Meas}(\Omega, \mathbb{R})$. We will also sometimes restrict ourselves to positive measurable functions (that is, $f : \Omega \rightarrow [0, \infty)$ measurable), and we will denote this set by $\text{Meas}^+(\Omega, \mathbb{R})$.

Since we will be working with preimages, the following result will be of great use.

Question 19. *Let $f : X \rightarrow Y$. Show that*

- (i) $f^{-1}(Y) = X$, $f^{-1}(\emptyset) = \emptyset$,
- (i) *if $S \subseteq Y$, then $f^{-1}(S^c) = f^{-1}(S)^c$, and*
- (ii) *if $\{S_n\}_{n=1}^\infty \subseteq Y$, then*

$$f^{-1}\left(\bigcup_{n=1}^\infty S_n\right) = \bigcup_{n=1}^\infty f^{-1}(S_n).$$

Use the above observations to show that if $\mathcal{S}$ is a family of subsets of Y and $\sigma(\mathcal{S})$ is the σ -algebra generated by $\mathcal{S}$, then

$$\sigma(f^{-1}(\mathcal{S})) = f^{-1}(\sigma(\mathcal{S})),$$

where $\sigma(f^{-1}(\mathcal{S}))$ is the σ -algebra generated by $f^{-1}(\mathcal{S})$.

The above allows us to greatly reduce our efforts and pick suitably nice sets to show measurability on. The following is a nice example.

Question 20. *Show that all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are Borel measurable. (Hint: It suffices to look at pre-images of open sets. Why?)*

Theorem 2.2. *If $(\Omega, \mathcal{F})$ is a measurable space, f, g measurable and $\lambda \in \mathbb{R}$ then we have*

- (i) λf is measurable
- (ii) $f + g$ is measurable
- (iii) fg is measurable
- (iv) $|f|$ is measurable
- (v) $\max\{f, g\}$ and $\min\{f, g\}$ are measurable.

Proof. We will show (ii) and leave the rest as an exercise for the reader which are a bit more direct.

We first show a naive approach which fails, in order to motivate a successful argument.

In order to show that $f + g$ is measurable, we need to show that for all Borel sets B , we have $(f + g)^{-1}(B) \in \mathcal{F}$. Writing out the set, we get

$$(f + g)^{-1}(B) = \{\omega \in \Omega \mid f(\omega) + g(\omega) \in B\}.$$

However, even in the case when B is a singleton, say $\{a\}$, we have $f(\omega) + g(\omega) = a$, which tells us nothing about $f(\omega)$ or $g(\omega)$. In principle, either one could be any real number, as long as the other compensates. With this observation, we may write (check this)

$$(f + g)^{-1}(B) = \bigcup_{r \in \mathbb{R}} f^{-1}(\{r\}) \cap g^{-1}(B - r).$$

Here $B - r = \{x \in \mathbb{R} | x = b - r \text{ for some } b \in B\}$ is a Borel set. (why?) However, if you are comfortable with σ -algebras, you might recognize that the above attempt doesn't quite work, as it is an uncountable union! If only we could make it countable somehow.

Luckily, we can make use of the properties of the Borel σ -algebra, in particular that they are generated by many families of *nice* sets (see Question 3). We argue that since the intervals (a, ∞) generate $\mathcal{B}(\mathbb{R})$, then it suffices to show that $(f + g)^{-1}((a, \infty)) \in \mathcal{F}$ for all $a \in \mathbb{R}$ to guarantee measurability. (why?)

With this strategy in mind, we can reduce our uncountable union down to a countable one with the following clever idea (check this equality):

$$(f + g)^{-1}((a, \infty)) = \bigcup_{k \in \mathbb{Z}} f^{-1}((-\infty, k]) \cap g^{-1}((a - k, \infty)).$$

By measurability of f and g and the closure under countable unions and intersections, we are thus guaranteed the measurability of $f + g$. $\square$

Question 21. Find a function f such that $|f|$ is measurable, but f is not.

Remark 2.3. Properties (i), (ii), and (iii) in the theorem above say that $\text{Meas}(\Omega, \mathbb{R})$ is an algebra. Property (v) guarantees us that it is also a lattice. So $\text{Meas}(\Omega, \mathbb{R})$ has a decently rich algebraic structure to it.

In addition, the question preceding the theorem above shows us that when $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then

$$C(\mathbb{R}, \mathbb{R}) \subseteq \text{Meas}(\mathbb{R}, \mathbb{R}).$$

This containment is strict. There is a much richer collection of measurable functions than continuous ones.

Example 2.4. A simple example of discontinuous measurable functions are the *characteristic functions*. Let $(\Omega, \mathcal{F})$ be a measurable space, and let $E \in \mathcal{F}$. Then define the associated characteristic function, χ_E as follows.

$$\chi_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E \\ 0 & \text{otherwise} \end{cases}.$$

It is easy to see that such a function is measurable. Indeed, let B be any Borel subset of $\mathbb{R}$, then we have

$$\chi_E^{-1}(B) = \begin{cases} \Omega & \text{if } 0, 1 \in B \\ E & \text{if } 1 \in B \text{ and } 0 \notin B \\ E^c & \text{if } 1 \notin B \text{ and } 0 \in B \\ \emptyset & \text{otherwise} \end{cases}.$$

So for example, if $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and $E = \mathbb{Q}$, then $\chi_{\mathbb{Q}}$, the characteristic function of the rational numbers, also known as the Dirichlet function, is everywhere discontinuous but measurable.

Curiosity 5. There are many questions we can ask about the nature of measurable functions to understand them better. For starters, what is $|\text{Meas}(\Omega, \mathbb{R})|$?

Also, how different are measurable functions from continuous functions, really? Though the Dirichlet function is discontinuous everywhere, it seems that if we changed its function values on a set of zero measure (the rationals), we could get a continuous function.

Does $\text{Meas}(\mathbb{R}, \mathbb{R})$ have any kind of natural metric or topology associated to it, to tell us when measurable functions are close? (We will soon see that in fact this space is closed under point-wise limits!)

Let's get on with defining our integrals. We proceed in stages, starting with the simplest functions, and using these to build up to the most general cases. For the following development, fix a measure space $(\Omega, \mathcal{F}, \mu)$.

Step One: Characteristic Functions

Let $E \in \mathcal{F}$ and consider its characteristic function χ_E . If we apply Lebesgue's approach to defining the integral, we come up with the following reasonable definition for the integral of E .

$$\int \chi_E d\mu := \mu(E).$$

So far so good, but characteristic functions are a very limited class of functions. We can improve this immediately by extending this definition linearly, as follows.

Step Two: Simple Functions

Definition 2.5. A measurable function $\varphi : \Omega \rightarrow \mathbb{R}$ is called *simple* if $|\text{Ran } \varphi| < \infty$.

Question 22. Show that simple functions are exactly the linear combinations of characteristic functions.

Let φ be simple with $\text{Ran } \varphi = \{r_1, \dots, r_n\}$. Then define

$$\int \varphi d\mu := \sum_{k=1}^n r_k \cdot \mu(\varphi^{-1}(\{r_k\})).$$

This is still giving us a seemingly very small class of functions to work with, but now that we have these simple functions, we can use these to approximate more arbitrary functions. We start with the positive ones.

Step Three: Positive Functions

Even with Riemann's approach, we approximated the area under a curve with simple functions which were constant on subintervals. Lebesgue's idea of chopping up the range allows us to use simple functions which are constant on measurable partitions, expanding the class of approximation functions significantly. But the following definition closely parallels the Riemann approach.

Definition 2.6. If $f : \Omega \rightarrow [0, \infty)$ is measurable define

$$\int f d\mu = \sup_{\substack{\varphi \text{ simple} \\ \varphi \leq f}} \int \varphi d\mu.$$

Question 23. Could we have defined this as an infimum instead, over all simple functions larger than f ?

Step Four: Generalizing

To extend this integral to a more general class of measurable functions, we use the ability to write a function in terms of positive ones. The following lemma takes care of this.

Lemma 2.7. If $f \in \text{Meas}(\Omega, \mathbb{R})$ then there exist $f_+, f_- \in \text{Meas}^+(\Omega, \mathbb{R})$ such that

$$f = f_+ - f_-$$

Proof.

□

Now one might naturally come up with the following extension of the integral to f :

$$\int f d\mu := \int f_+ d\mu - \int f_- d\mu.$$

This is indeed how we do this, however we offer one caveat. Namely, the integrals may not have been finite, so there is potential in the above that we are left with the paradoxical

$$\int f d\mu = \infty - \infty.$$

To avoid this, we simply stipulate that for the integral of f to be well defined, then at most one of $\int f_+ d\mu$ and $\int f_- d\mu$ is infinite.

Definition 2.8. We will say a measurable function f is *integrable* if

$$\int |f| d\mu = \int f_+ d\mu + \int f_- d\mu < \infty.$$

Denote by $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ the space of integrable functions on $(\Omega, \mathcal{F}, \mu)$.

Now that the integral is properly defined, let's take a moment to verify some basic properties. The following lemma says that the integral is both *monotone* and *linear*.

Lemma 2.9. If f, g are integrable with $f \leq g$, then

$$\int f d\mu \leq \int g d\mu.$$

Furthermore, if $\lambda \in \mathbb{R}$, we have

$$\int (\lambda f + g) d\mu = \lambda \int f d\mu + \int g d\mu.$$

The Lebesgue integral respects limits very nicely too. We start with measurability.

Theorem 2.10. If $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of measurable functions which converges point-wise to f , then f is measurable.

Theorem 2.11 (Monotone Convergence Theorem). If $\{f_n\}_{n \in \mathbb{N}}$ is a family of non-negative, measurable functions such that $f_{n+1} \geq f_n$ and $f(x) := \sup_{n \in \mathbb{N}} f_n(x)$, then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

This discussion has been largely abstract. Let's see if we can compute some examples!

Example 2.12. Let Ω be any set and let $\mathcal{F} = \mathcal{P}(\Omega)$. Fix $a \in \Omega$ and consider the Dirac mass δ_a . Then integration against δ_a has a surprising form: for any $f \in \text{Meas}(\Omega, \mathbb{R})$, we have

$$\int f d\delta_a = f(a).$$

So integration of a function against the Dirac mass at a is just evaluation of f at a !

This is not too difficult to show directly from the definition, see if you can.

Example 2.13. Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, and take the counting measure μ_{count} . Then, for $f : \mathbb{N} \rightarrow \mathbb{R}$ integrable with respect to μ_{count} , we have

$$\int f d\mu_{\text{count}} = \sum_{n=1}^{\infty} f(n).$$

Furthermore, f is integrable with respect to μ_{count} if and only if its corresponding series is absolutely summable.

So every time you evaluated a function or took a summation, you were doing integrals!

2.2. Exercises.

Question 24. Show that if $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then it is Lebesgue integrable, and furthermore its Lebesgue integral equals its Riemann integral.

See if you can generalize this to all Riemann integrable functions on $[a, b]$.

Question 25. Show that $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ is a vector space equipped with a semi-norm

$$\|f\|_1 = \int |f| d\mu.$$

Why is it not a norm in general?

When taking a quotient of $\mathcal{L}^1$ by the relation $f \sim g$ if $\|f - g\|_1 = 0$, we get the quotient space

$$L^1(\Omega, \mathcal{F}, \mu) = \mathcal{L}^1(\Omega, \mathcal{F}, \mu) / \sim.$$

It is a nice fact that this turns out to be a Banach space.

3. LECTURE 3: THE RIESZ-MARKOV-KAKUTANI REPRESENTATION THEOREM

Coming soon!

One way to think about measures is as distributions of stuff (e.g. mass or energy) throughout the underlying space. To get a feel for the distribution then, we can look to certain functions to test the distribution with.

For example, suppose that our measure gives us the temperature (density of heat) throughout a space. Using a thermometer to probe the space at various points gives us a feel for how the heat is distributed. But thermometers don't allow us to evaluate heat at specific points. Rather they tend to average out the heat

Let's look at a few examples.

Example 3.1. The simplest measures are the Dirac masses. Here the mass is concentrated at a single point. Regardless of how we probe this space.

4. LECTURE 4: RADON-NIKODYM DERIVATIVES

Given a measure space $(\Omega, \mathcal{F}, \mu)$ and a measurable function $f : \Omega \rightarrow \mathbb{R}$, we can *push forward* μ to a measure on the real line. For each Borel set B , define

$$\mu_f(B) := \mu(f^{-1}(B)).$$

It is straightforward to verify that this is a Borel measure on $\mathbb{R}$ due to the measurability of f .

In probability theory (see Supplement 1 below), we start with a *probability space* $(\Omega, \mathcal{F}, P)$, where P is just a measure with the added constraint that $P(\Omega) = 1$. Measurable functions are then called *random variables*, and the pushforward measure described above is called the *distribution* of the random variable f .

Question 26. *Do every two different measurable functions yield different pushforward measures on $\mathbb{R}$?*

The pushforward allows us to turn an integral over our sample space Ω into an integral over $\mathbb{R}$.

Question 27. *Show that*

$$\int_{\Omega} f(\omega) d\mu(\omega) = \int_{\mathbb{R}} x d\mu_f(x).$$

In probability theory, f is called a random variable and then the quantity above is the expected value of f .

As you may have noticed, this transformation wasn't too profound. However, we also often perform another little trick. Suppose for example that f has a *standard normal distribution*. Then

$$\int_{\Omega} f(\omega) dP(\omega) = \int_{\mathbb{R}} x d\mu_f(x) = \int_{\mathbb{R}} x \left(\frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \right) dx.$$

Here dx refers to the Lebesgue measure on $\mathbb{R}$. This change in the measure of integration is useful enough to appear in most expected value calculations.

It would be nice if we could always just convert back to a nice measure like the Lebesgue measure to perform our calculations and analysis. However, can we really do this all the time?

Our hope dies very quickly when considering a simple measure like a Dirac mass. For example we saw in Lecture 2 that for any measurable f ,

$$\int_{\mathbb{R}} f(x) d\delta_a = f(a).$$

So in particular,

$$\int_{\mathbb{R}} \chi_{\{a\}} d\delta_a = 1.$$

But if we try to change this to an integral against the Lebesgue measure, we run into problems. Note that $\chi_{\{a\}} f(x) = f(a) \chi_{\{a\}}$, so that

$$\int_{\mathbb{R}} \chi_{\{a\}} f(x) dx = f(a) \int_{\mathbb{R}} \chi_{\{a\}} dx = f(a) \lambda(\{a\}) = 0.$$

We thus arrive at a contradiction if we suppose that we can find some $f(x)$ such that

$$\int_{\mathbb{R}} g(x) d\delta_a = \int_{\mathbb{R}} g(x) f(x) dx$$

for all integrable $g(x)$.

What went wrong in the above counter-example was the existence of a set $(\{a\})$ which was massive for δ_a , but massless for λ . The existence of such sets yields contradictions in a very similar way to what is seen above.

In general, if μ_1 and μ_2 are such that we could find an f with

$$\int_{\Omega} g d\mu_1 = \int_{\Omega} g f d\mu_2,$$

for all g , then by choosing $g = \chi_E$, we see that it must be the case that whenever $\mu_2(E) = 0$, it follows that $\mu_1(E) = 0$ (check this).

This leads us to the following definition.

Definition 4.1. Given two measures μ_1 and μ_2 on $(\Omega, \mathcal{F})$, we say that μ_1 is *absolutely continuous with respect to* μ_2 , and write $\mu_1 \ll \mu_2$ if whenever $S \in \mathcal{F}$ with $\mu_2(S) = 0$, it follows that $\mu_1(S) = 0$.

Example 4.2. Motivated by considerations above, it is easy to see that for any $a \in \mathbb{R}$,

$$\lambda \ll \delta_a \quad \text{but} \quad \delta_a \not\ll \lambda.$$

Curiosity 6. One might ask the question, is there a measure, μ , on Ω such that for every other measure ν , we have $\nu \ll \mu$? What about $\mu \ll \nu$?

For the first, we note that if $\mu \gg \nu$ for every measure ν , then, for example $\mu \gg \delta_a$ for every $a \in \Omega$. So in particular, we must have that $\mu(\{a\}) > 0$ for every $a \in \Omega$. One such measure for which this holds is the counting measure! Can you show that every other measure on Ω is absolutely continuous with respect to the counting measure?

For the second, we may note simply that the zero measure is zero on every subset. So if $\mu \ll \nu$ for every measure, then $\mu \ll 0$, which forces us to conclude that μ is in fact the zero measure. It is immediate that the zero measure also satisfies the property of being absolutely continuous with respect to every measure.

Remark 4.3. We note that while absolute continuity is transitive, it is not anti-symmetric. That is, we can have two different measures μ_1 and μ_2 such that $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_1$. One simple example of this is to take any measure μ_1 and let $\mu_2 = c\mu_1$ for any $c > 0$.

In this case, since we can select $c > 1$ or $c < 1$, we see that the relation $\ll$ is not so much about one measure being larger than another in a “point-wise” sense. Rather, it's more of a comparison of the relative *supports* of the two measures (i.e. on which sets the mass lives).

What is remarkable, is that if our measures are relatively well behaved (σ -finite), this condition of absolute continuity is all we need! The following is the main result of this lecture.

Theorem 4.4 (Radon-Nikodym Theorem). If μ_1 and μ_2 are σ -finite measures on $(\Omega, \mathcal{F})$ such that $\mu_1 \ll \mu_2$ then there exists a measurable $f : \Omega \rightarrow [0, \infty)$ such that for all $E \in \mathcal{F}$,

$$\mu_1(E) = \int_E f \, d\mu_2.$$

Furthermore, if $g : \Omega \rightarrow [0, \infty)$ is measurable such that $\mu_1(E) = \int_E g \, d\mu_2$, then $g = f$ almost everywhere.

See Question 29 below for an example of how the Radon-Nikodym theorem fails when we aren't dealing with σ -finite measures.

Question 28. This might not look like what is promised, however, show that the above implies that

$$\int_{\Omega} g d\mu_1 = \int_{\Omega} g f d\mu_2.$$

(Hint: Build up to f through characteristic functions, then simple functions, then limits as in the development of the integral.)

4.1. Exercises.

Question 29. We saw that every measure is absolutely continuous with respect to the counting measure μ_{count} . Use this fact about the Lebesgue measure to conclude that the Radon-Nikodym theorem is not necessarily true when our measures are not σ -finite.

Question 30. Recall from Question 16 (Lecture 1), that every σ -finite measure μ on $\mathbb{N}$ is determined by a weight function $w : \mathbb{N} \rightarrow [0, \infty)$. Furthermore, we saw that every measure is absolutely continuous with respect to the counting measure, which on $\mathbb{N}$, is σ -finite itself. Show that the Radon-Nikodym derivative of μ with respect to the counting measure is exactly the weight function w .

5. LECTURE 5: SIGNED MEASURES

One idea that may have sprung to mind during the discussion of the Radon-Nikodym derivatives of the last lecture, was “what if the derivative function f was an arbitrary integrable function?”. In particular, what happens if we define

$$\nu(E) = \int_E f \, d\mu$$

where now $f : \Omega \rightarrow \mathbb{R}$ is simply integrable, not necessarily non-negative?

On one hand we might easily dismiss this, as, for example, we could let $E = f^{-1}((-\infty, 0))$, and then we would have to conclude that if $\mu(E) > 0$, then

$$\nu(E) < 0.$$

However, this object ν still behaves like a measure in the sense that $\nu(\emptyset) = 0$, and ν is countably additive.

Question 31. *Show that ν , defined as above, is countably additive. You will need a limit theorem for integrals for this.*

So ν is a measure, that measures some sets as negative. We call such an object a *signed measure*. More formally,

Definition 5.1. A set function $\mu : \mathcal{F} \rightarrow [-\infty, \infty)$ or $\mu : \mathcal{F} \rightarrow (-\infty, \infty]$ a *signed measure* if

- (1) $\mu(\emptyset) = 0$, and
- (2) whenever $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

Remark 5.2. In the definition, we disallow μ to take on both positive and negative infinite values. For if E and F are pairwise disjoint such that $\mu(E) = \infty$ and $\mu(F) = -\infty$, then it is not clear how we compute

$$\mu(E \cup F) = \mu(E) + \mu(F) = \infty - \infty = ?$$

Signed measures might seem like generalization for the sake of generalization, however they seem to naturally sit in some gaps left behind by our discussions of positive measures. For example, from Question 5 in Lecture 1, you may have noticed that only the *positive* linear combinations of measures are still measures. This left us to work in the setting of a convex cone, not a vector space. It would be nice to work in a vector space for added structure! Luckily signed measures rectify this.

Indeed, if μ_1 and μ_2 are two (positive) measures on Ω such that $\mu_1(\Omega) < \infty$ or $\mu_2(\Omega) < \infty$ then it is easy to see that *any* linear combination of these two measures is a signed measure. So in particular, restricting to the linear combinations of finite (positive) measures yields a vector space!

Question 32. *What goes wrong if we allow both $\mu_1(\Omega) = \infty$ and $\mu_2(\Omega) = \infty$?*

So we can get signed measures from linear combinations of positive measures, but do we get all of them this way? That is, if we start with a signed measure, can we necessarily decompose it into positive measures?

Let μ be a signed measure on Ω . If we are to decompose it into positive measures, we need to account for all of the “negative mass” of μ by scaling a positive measure somehow. However we run into an immediate hurdle when we try to do this naively. In particular, it does not suffice to simply take an arbitrary set with total mass positive, as the exercise below shows.

Question 33. *Find a signed measure μ , a set P , and a subset $E \subseteq P$ such that $\mu(P) > 0$ and $\mu(E) < 0$.*

This is an annoyance that makes decomposing μ into positive measures seemingly impossible at first glance. If we were to do so, we would really need to find sets where the mass is *purely* positive or negative. This leads us to the following definition:

Definition 5.3. Let μ be a signed measure on $(\Omega, \mathcal{F})$. We say a subset $P \in \mathcal{F}$ is a *positive set* (for μ) if whenever $E \subseteq P$, we have $\mu(E) \geq 0$. Similarly, we say that N is a *negative set* (for μ) if whenever $E \subseteq N$ we have $\mu(E) \leq 0$.

Our decomposition problem is notably eased if we can thus corral the negative mass into one large negative set and the positive mass into one large positive set.

Question 34. Show that countable unions of positive sets are positive, and countable unions of negative sets are negative. Show that an uncountable union of positive sets need not be positive.

It's not immediately obvious that we can even find such sets in the first place. But the following lemma guarantees this. The idea is to take a set of positive total mass, and keep cutting away subsets of negative mass from it.

Lemma 5.4. Let μ be a signed measure on $(\Omega, \mathcal{F})$. Suppose that for some $S \in \mathcal{F}$, we have $\mu(S) > 0$. Then there exists a positive $P \subseteq S$ such that $\mu(P) > 0$.

Proof. Let μ be a signed measure on $(\Omega, \mathcal{F})$. Without loss of generality, we can consider μ such that $\mu : \mathcal{F} \rightarrow [-\infty, \infty)$ (e.g. by multiplying μ by -1 if necessary). Fix $S \subseteq \Omega$ such that $\mu(S) > 0$.

The idea will be that if S is not positive, we can successively cut out subsets with a negative mass, leaving us with only a positive set remaining. With this line of thinking, let's record how much negative mass could exist in S by letting

$$\alpha_0 = \inf\{\mu(E) : E \subseteq S\} < 0.$$

A priori, it could be the case that $\alpha_0 = -\infty$ so let's cut out negative mass bit by bit.

Pick $E_1 \subseteq S$ such that $\mu(E_1) \leq \max\{-1, \frac{\alpha_0}{2}\}$. If we let $S_1 = S \setminus E_1$, then we see that $S_1 \subseteq S$ and

$$\mu(S_1) = \mu(S) - \mu(E_1) = \mu(S) + |\mu(E_1)|.$$

So if S_1 is positive, we are done. Otherwise we can let $\alpha_1 = \inf\{\mu(E) : E \subseteq S_1\} < 0$. Then find $E_2 \subseteq S_1$ such that $\mu(E_2) \leq \max\{-1, \frac{\alpha_1}{2}\}$.

Continue this process recursively, by letting

$$S_n = S_{n-1} \setminus E_n = S \setminus \left(\bigcup_{k=1}^n E_k \right),$$

selecting $E_{n+1} \subseteq S_n$ such that

$$\mu(E_n) \leq \max\left\{-1, \frac{\inf\{\mu(E) : E \subseteq S_n\}}{2}\right\}.$$

For simplicity, rather than terminating our construction upon hitting a positive set, we will keep pulling out sets of measure zero for every natural number n in the above way (note that E_n may be empty).

Then let

$$P = \bigcap_{n=1}^{\infty} S_n = S \setminus \left(\bigcup_{n=1}^{\infty} E_n \right)$$

and we claim that P is a positive set. First thing to note is by σ -additivity,

$$\mu(P) = \mu(S) - \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu(S) - \sum_{n=1}^{\infty} \mu(E_n) \geq \mu(S) > 0$$

From this observation, it is immediately deduced that $\mu(E_n) \rightarrow 0$, as otherwise, we reach the conclusion that $\mu(P) = \infty$, which contradicts our assumption that $\mu : \mathcal{F} \rightarrow [-\infty, 0)$.

Lastly, to show that P is a positive set, let $F \subseteq P$, then $F \subseteq S_n$ for all $n \in \mathbb{N}$. So in particular, for every $n \in \mathbb{N}$ we have

$$\mu(F) > \mu(E_n) \rightarrow 0$$

guaranteeing us that $\mu(F) \geq 0$. □

Theorem 5.5 (Hahn Decomposition Theorem). *For any signed measure μ , there exists a positive set P and negative set N such that $P \cup N = \Omega$.*

This theorem is largely a consequence of the previous lemma. Its proof is now relatively short and sweet.

Proof. Let μ be a signed measure on $(\Omega, \mathcal{F})$. We try to build the *largest* positive set P as follows. Let

$$\alpha = \sup\{\mu(S) : S \text{ is positive}\}.$$

Note that $\alpha \geq 0$ since $S = \emptyset$ is a positive set.

Next, for each $n \in \mathbb{N}$, let P_n be a positive set with $\mu(P_n) > \alpha - \frac{1}{n}$. Define

$$P := \bigcup_{n=1}^{\infty} P_n \quad \text{and} \quad N := \Omega \setminus P.$$

It is quick to see that P is a positive set. In fact, $\mu(P) = \alpha$. To see this, note that since P is positive, for any $S \subseteq P$, we have $\mu(S) \leq \mu(P)$ (as $\mu(P) = \mu(S) + \mu(P \setminus S)$). So for any $n \in \mathbb{N}$,

$$\alpha - \frac{1}{n} < \mu(P_n) \leq \mu(P) \leq \alpha,$$

yielding $\mu(P) = \alpha$.

Lastly, we will show that N is a negative set. Indeed, suppose N was not negative. Then there would be some $E \subseteq N$ such that $\mu(E) > 0$. However, by the lemma above, we must conclude that there is a positive set $P' \subseteq E$ such that $\mu(P') > 0$. But then $P \cup P'$ is a positive set and a disjoint union, so that

$$\mu(P \cup P') = \mu(P) + \mu(P') > \mu(P) = \alpha,$$

a contradiction. □

In fact, the above decomposition has a certain character of uniqueness to it.

Question 35. *Show that if (P, N) and (P', N') are two Hahn decompositions of Ω (for μ), then $\mu(P \Delta P') = 0$ and hence $\mu(N \Delta N') = 0$ where Δ refers to the symmetric difference of the sets.*

Note that this is saying that any other Hahn decompositions do not carry any difference in mass. That is, we have uniqueness up to measure zero sets.

Now we can finally get to our decomposition of signed measures into positive ones.

Theorem 5.6 (Jordan Decomposition Theorem). *For any signed measure μ on $(\Omega, \mathcal{F})$, there are positive measures μ_+ and μ_- , at least one of which is finite, such that*

$$\mu = \mu_+ - \mu_-$$

and this decomposition is unique.

The proof is a nice, quick, and accessible puzzle which rides on the power of the Hahn Decomposition Theorem.

Question 36. *Prove the Jordan Decomposition Theorem.*

So we have succeeded in our goal to write any signed measure in terms of positive measures.

5.1. Complex Valued Measures. The above discussion opens the door for $\mathbb{C}$ -valued measures as well. However dealing with infinities is no longer tenable in this case.

Definition 5.7. A $\mathbb{C}$ -valued measure on $(\Omega, \mathcal{F})$ is a map $\mu : \mathcal{F} \rightarrow \mathbb{C}$ satisfying

- $\mu(\emptyset) = 0$, and
- If $\{A_n\}_{n \in \mathbb{N}}$ is a collection of pairwise disjoint subsets in $\mathcal{F}$, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n).$$

We might naturally ask if we can understand these too in terms of positive measures. Luckily we have done all of our work already. Just need one simple idea from complex variables.

In particular, let

$$\mu_r = \frac{\mu + \bar{\mu}}{2} \quad \text{and} \quad \mu_i = \frac{\mu - \bar{\mu}}{2i}.$$

Then it is easy to see that μ_r and μ_i are (real-valued) signed measures. So we can decompose $\mu_r = \mu_{r,+} - \mu_{r,-}$ and $\mu_i = \mu_{i,+} - \mu_{i,-}$. This leaves us with the decomposition

$$\mu = \mu_r + i\mu_i = \mu_{r,+} - \mu_{r,-} + i\mu_{i,+} - i\mu_{i,-}.$$

Now that we have a complex vector space structure on the set of measures, we might wonder if there is any natural norm or topology on this space. Indeed, by looking at the absolute value measure, we get a natural norm:

Definition 5.8. Given a complex measure μ , its *total variation* is the measure defined by

$$|\mu|(E) = \mu_{r,+}(E) + \mu_{r,-}(E) + \mu_{i,+}(E) + \mu_{i,-}(E).$$

Furthermore, the corresponding *total variation norm* of μ is the quantity

$$\|\mu\| = |\mu|(\Omega).$$

Question 37. Show that $\|\mu\|$ is a norm on the set of complex measures. Show furthermore that it is in fact complete, giving us a Banach space.

There are other natural topologies to consider on this space, motivated by our discussions in Lecture 3. In particular, it still makes good sense, due to the decomposition into positive measures, to integrate against a complex measure μ . In particular, for $f : \Omega \rightarrow \mathbb{C}$ integrable, define

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f \, d\mu_{r,+} - \int_{\Omega} f \, d\mu_{r,-} + i \int_{\Omega} f \, d\mu_{i,+} - i \int_{\Omega} f \, d\mu_{i,-}.$$

Then it follows that each such μ naturally defines a functional on the set of compactly supported continuous functions from a locally compact Hausdorff space.

Question 38. Can we extend the Riesz-Markov-Kakutani representation theorem to complex measures? Are all linear functionals decomposable into positive ones?

5.2. Exercises.

Question 39. Show that if μ is a (positive) measure on $(\Omega, \mathcal{F})$ and $f : \Omega \rightarrow \mathbb{R}$ is integrable, then the measure ν defined by

$$\nu(E) = \int_E f \, d\mu$$

is a signed measure. Furthermore, find a more explicit expression for ν_+ and ν_- (in terms of f).

Question 40. It is not the case that every complex linear combination of signed measures is a $\mathbb{C}$ -valued measure. What restrictions do we need on the set of signed measures to recover the $\mathbb{C}$ -valued measures via linear combinations?

6. LECTURE 6: PRODUCT MEASURES

We've encountered the length area in

Question 41. *Show that $\mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$.*

7. LECTURE 7: THE FUNDAMENTAL THEOREM OF CALCULUS

Since we have a refined notion of integral from Lebesgue, we might wonder to what extent we can carry over the cherished results of the Riemann integral, such as the fundamental theorems of calculus. The Lebesgue integral seems to be able to handle a larger class of functions, so perhaps we get a nice generalization here.

Indeed, one potential point of contact is that part 1 of the fundamental theorem of calculus demands that we start with a continuous integrand, only then can we guarantee a differentiable accumulation function. However, there are many more even Riemann integrable functions than just the continuous ones. So maybe we if we find the right relaxation of differentiability and approach things with the Lebesgue measure, we can get a sufficiently satisfying conclusion.

7.1. Positive Integrals and Increasing Functions. As alluded to above, we will start with addressing the question of *what types of functions are accumulation functions corresponding to Lebesgue integrable functions?* Furthermore, the simplest place to start here would be to work with the positive integrables.

Its fairly immediate to see that if $f(x)$ is positive and integrable on $[a, b]$ then its accumulation function,

$$F(x) = \int_a^x f(x)dx,$$

is non-decreasing. Furthermore, we can squeeze a bit more juice from this lemon by suggesting that $f(x)$ only needs to be positive on a set of *full measure*. That is, we can allow f to be negative or even undefined on a set of measure zero, which has no impact on $F(x)$.

Okay, that was a pretty simple observation. This leads us to the reverse question: *are all non-decreasing functions $F(x)$ differentiable with non-negative derivative, at least on a set of full measure?* It seems like this question may be to naive to hope for a positive answer, yet somewhat miraculously, the answer is yes!

Theorem 7.1 (Lebesgue Differentiation Theorem). *If $F : [a, b] \rightarrow \mathbb{R}$ is non-decreasing, then F is differentiable λ -a.e. Furthermore, its derivative F' is non-negative λ -a.e., measurable, and*

$$\int_a^b F'(x)d\lambda \leq F(b) - F(a).$$

There is a lot to unpack here. Let's start by addressing some concerns. One might worry, for example, that an arbitrary non-decreasing function need not be even continuous, let alone differentiable! However, part of the power of the Lebesgue integral that lends itself to this fantastic result, is that we may ignore sets of measure zero. For example, it is relatively simple to argue that non-decreasing functions may only be discontinuous on a countable set.

Question 42. *Show that if $F : [a, b] \rightarrow \mathbb{R}$ is non-decreasing, then it's set of discontinuities is at most countably infinite. Thus its set of discontinuities has Lebesgue measure zero.*

Still, differentiability is a much more powerful property. This is true, and reflected in the highly non-trivial proof of the Differentiation Theorem. The proof will be handled later in bits and pieces, with one major technical result (the Vitali Covering Lemma) left as an appendix. First, let's take a quick look at what it means to be differentiable almost everywhere.

Question 43. *Show that $\chi_{[a,b]}$ is differentiable almost everywhere. What is it's derivative? Deduce that linear combinations of these are differentiable almost everywhere.*

Are all characteristic functions differentiable almost everywhere?

Example 7.2. Cantor Function

Okay then, what's the deal with the inequality in the statement? The fundamental theorem of calculus would have us believe this should be an equality. However, from our example of the Cantor function above, we see that we must let this go.

7.2. Functions of Bounded Variation.

8. SUPPLEMENT 1: PROBABILITY THEORY

Measure theory is the formal basis of probability theory where we only make one additional assumption.

Definition 8.1. A measure space $(\Omega, \mathcal{F}, \mu)$ is called a *probability space* if $\mu(\Omega) = 1$.

Note by $\text{Prob}(\Omega, \mathcal{F})$ the set of probability measures on $\mathcal{F}$.

Remark 8.2. This additional assumption on the measure for probability spaces means that naively taking positive linear combinations of probability measures need not yield a probability measure. However, some such combinations still do. In particular, the *convex* combinations.

Question 44. Show that convex combinations of probability measures are probability measures. That is, $\text{Prob}(\Omega, \mathcal{F})$ is a convex subset of $\mathcal{M}(\Omega, \mathcal{F})$.

Curiosity 7. So the set of probability measures on Ω is guaranteed to be convex. One natural place to look next is for extreme points. These are the points that cannot be written as a non-trivial convex combination of two other distinct points. For example, a rectangle, including its inner area is convex with its vertices as its extreme points. This leads us to the question: what are the extreme points of $\text{Prob}(\Omega, \mathcal{F})$? Maybe the case when Ω is finite is more tractable?

In probability theory, we are usually concerned with studying random variables. However, we have also already seen these, as random variables are simple measurable functions!

Definition 8.3. Given a probability space $(\Omega, \mathcal{F}, P)$, a *random variable* is a (Borel) measurable function $f : \Omega \rightarrow \mathbb{R}$.

The “randomness” in a random variable comes from ignoring their domain and instead looking at the “distribution” of their outputs. Even though a function is by definition deterministic, if we blind ourselves to the inputs of the function, then the outputs will appear “random” (but following a distribution).

Definition 8.4. Given a random variable $X : \Omega \rightarrow \mathbb{R}$, the *distribution of X* is the (probability) measure $\mu_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ defined by

$$\mu_X(B) := P(X^{-1}(B)), \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

Remark 8.5. So in particular, distributions of random variables are actually just (Borel) probability measures on $\mathbb{R}$! It is not unfair to say that studying probability theory is studying the space $\text{Prob}(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In fact, this space has certain algebraic structure to it, in the form of *convolution* (the distribution you get when you add two independent random variables with corresponding distributions). We defer a closer look into this until later.

Associated to distributions are the *distribution functions*. However, these are simply properties of measures on $\mathbb{R}$.

Definition 8.6. For $\mu \in \mathcal{M}(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, define its *cumulative distribution function*, $F_\mu : \mathbb{R} \rightarrow [0, \infty]$ by

$$F_\mu(x) := \mu((-\infty, x]).$$

Fact 1. The density function associated to μ is the Radon-Nikodym derivative of the measure μ with respect to the Lebesgue measure. (See Lecture 4.) As a consequence, this does not exist when μ has atoms (i.e. points $x \in \mathbb{R}$ such that $\mu(\{x\}) > 0$), but often does exist even when the cumulative distribution function F_μ is not differentiable!

Of course, when F_μ is differentiable, then the density function f_μ is simply the derivative of F_μ .

The next thing one picks up in an introductory probability course is the notion of expected value for a random variable. If not introduced via measure theory, one might have to split our definition up into a case for discrete random variables and a case for continuous random variables. However, with the Lebesgue integral, we no longer need to make this distinction.

Definition 8.7. Given a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, its *expected value* is the quantity

$$\mathbb{E}(X) := \int_{\Omega} X dP = \int_{\mathbb{R}} x d\mu_X.$$

Question 45. *Verify that the equality in the definition above is true.*

Question 46. *Show that if $\text{Ran } X = \{1, 2, 3\}$ and*

$$P(X^{-1}(\{1\})) = p_1, \quad P(X^{-1}(\{2\})) = p_2, \quad \text{and} \quad P(X^{-1}(\{3\})) = p_3,$$

then

$$\mathbb{E}(X) = 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3.$$

9. SUPPLEMENT 2: L^p SPACES

9.1. L^∞ . When $p = \infty$ we get a space that behaves differently than the rest.

Definition 9.1. For a measure space $(\Omega, \mathcal{F}, \mu)$ and a measurable function $f : \Omega \rightarrow \mathbb{C}$, we define its *essential supremum* to be the quantity

$$\|f\|_{\infty, \mu} = \inf\{c \mid \mu(\{|f|(\omega) \geq c\}) = 0\}.$$

Furthermore we let

$$\mathcal{L}^\infty(\mu) = \{f : \Omega \rightarrow \mathbb{C} \mid \|f\|_{\infty, \mu} < \infty\}.$$

This is very much like a sup-norm, except that we ignore sets of measure zero.

Question 47. Find an example of a Borel measurable function $f : \mathbb{R} \rightarrow \mathbb{C}$ such that $\|f\|_\infty = \infty$ but $\|f\|_{\infty, \lambda} = 0$, where λ is the Lebesgue measure on $\mathbb{R}$. Conclude that $\|\cdot\|_{\infty, \lambda}$ is not a norm on $\mathcal{L}^\infty(\mu)$.

Question 48. Given $a \in \Omega$, find a simpler expression for $\|f\|_{\infty, \delta_a}$.

As we saw above, and in much the same way as we consider in the $\mathcal{L}^p$ spaces, the space $\mathcal{L}^\infty(\mu)$ can be too large for $\|\cdot\|_{\infty, \mu}$ to be a true norm. However, we can rid ourselves of this problem by taking a quotient. Define

$$L^\infty(\mu) := \mathcal{L}^\infty(\mu) / \{f \mid \|f\|_{\infty, \mu} = 0\}.$$

Question 49. Show that $L^\infty(\mu)$ is a Banach space. Furthermore, show that it is separable if and only if $\mu = \sum_{k=1}^n c_k \delta_{a_k}$ (so most L^∞ spaces are inseparable).

Curiosity 8. We have an L^∞ space for every measure μ , but it is reasonable to ask how they relate to each other. Are they all different?